

Nebula9 Enterprise Knowledge Intelligence Reference Platform

Reusable, industry-configurable demonstration platform for enterprise knowledge discovery, multi-agent reasoning, evidence validation and governed AI answers.

Document purpose	Define the build scope, architecture, controls, evaluation and sales-demo requirements for a reusable reference platform.
First implementation	Manufacturing / Industrial Knowledge Intelligence
Future configurations	Retail & CPG, Technology & Services, Insurance, Healthcare & Life Sciences, Financial Services
Primary audience	Nebula9 leadership, CXO prospects, solution architects, engineering team and project intern
Product posture	Demo/reference platform built with production-grade architecture and a clean path to production
Development model	AI-assisted development permitted; human ownership, review, testing and security verification required

Core proposition

Do not build another generic chatbot. Demonstrate how an enterprise can ask one business question across disconnected knowledge sources, have specialized AI agents retrieve and reason over the right information, validate evidence, and return a traceable answer.

Source basis: This PRD carries forward the source case study's core framing: databases, documents and controlled external sources; specialized agents; one consolidated contextual response; evidence-backed and source-traceable answers; modular/reusable architecture; and enterprise-oriented governance principles. fileciteturn1file0

1. Executive Summary

Nebula9 previously built a Multi-Agent RAG capability for enterprise environments. Because the underlying client implementations are covered by NDA, the objective of this project is not to expose, copy or recreate any client-specific implementation or data. The objective is to create a sanitized, synthetic/public-data reference implementation that demonstrates the same class of capability in a reusable and sales-ready form.

The platform should be positioned externally as Enterprise Knowledge Intelligence. “Multi-Agent RAG” is an implementation mechanism, not the primary sales proposition.

The first complete implementation should use a Manufacturing / Industrial scenario because it naturally supports multiple knowledge types—operational documents, structured production/quality data and controlled external intelligence—without making the first demo primarily about sector-specific regulatory compliance. Retail & CPG should be validated as the second industry configuration to prove that the architecture is genuinely reusable.

The resulting application should allow a CXO to see a realistic business problem, ask a complex question, observe the platform retrieve information from multiple source types, understand which specialized agents participated, inspect evidence and source traceability, see how conflicts or insufficient evidence are handled, and then understand how the same architecture can be configured for another industry.

2. Product Vision & Positioning

2.1 Product hierarchy

1. Business proposition	Enterprise Knowledge Intelligence — turn fragmented enterprise knowledge into trusted, contextual answers.
2. Core platform	Multi-agent orchestration + retrieval + structured data access + controlled external intelligence + evidence validation + synthesis.
3. Enterprise controls	Authentication, RBAC, source/workspace access, auditability, governance, observability and security controls.
4. Industry configuration	Manufacturing, Retail & CPG, Technology & Services, Insurance, Healthcare & Life Sciences, Financial Services.
5. Business use case	Operations, supply chain, quality, research, risk, compliance, customer intelligence, decision support and other domain-specific scenarios.

2.2 What the product is

- A reusable enterprise knowledge intelligence reference platform.
- A demonstration of specialized AI agents working together across heterogeneous sources. • An evidence-first answer experience with source traceability.
- A configurable foundation where industry data, terminology, scenarios, source connectors and evaluation sets can change without rebuilding the core application.
- A credible technical foundation that can be extended into a production engagement. **2.3**

What the product is not

- Not a generic conversational chatbot.
- Not a client-specific reproduction of an NDA implementation.
- Not a claim that the demo is production-ready for every enterprise environment or regulatory jurisdiction. • Not an excuse to hard-code industry behavior into the core codebase.
- Not a collection of AI-generated code that has not been understood, reviewed and tested.

3. Business Problem

The source case study identifies the underlying enterprise problem as fragmented knowledge across disconnected systems: teams manually combine information, research takes too long, there is no single view, decisions wait for research, and answer sources are difficult to verify. fileciteturn1file0

3.1 Demonstration problem

The demo must make this problem tangible. A business leader should be able to ask a question that cannot be answered reliably from a single document or a single database.

3.2 Manufacturing first scenario

- Unstructured knowledge: SOPs, maintenance manuals, quality reports, supplier documents, engineering notes, inspection reports, safety procedures and production reports.
- Structured knowledge: production volumes, downtime, quality metrics, inventory, supplier performance, procurement and cost data.
- Controlled external intelligence: approved public supplier/market/regulatory information or synthetic external sources.
- Example CXO question: “What are the major factors contributing to production downtime at Plant A, and what actions have previously reduced similar downtime?”
- The question should require multiple sources and demonstrate evidence, synthesis and traceability rather than simply semantic search.

4. Objectives & Success Criteria

4.1 Objectives

- Build a working end-to-end Enterprise Knowledge Intelligence demo.
- Demonstrate multi-agent orchestration across document, structured-data and controlled external sources.
- Make evidence and source traceability visible in the user experience.
- Implement security, access control, audit and observability as real platform capabilities—not just statements in documentation.
- Separate configurable industry behavior from core platform code.
- Validate portability by switching from Manufacturing to at least one second industry pack without rebuilding the platform.
- Create a polished CXO demo, demo script, architecture diagram and technical overview suitable for prospect conversations.

4.2 CXO Test

A non-technical executive should understand within 60–90 seconds: What problem does this solve? What information can it connect? What can I ask? What evidence will I receive? The first screen and demo flow must communicate these points without requiring knowledge of RAG, embeddings, vector databases or agent frameworks.

4.3 Definition of success

The project succeeds when a prospect can see a realistic business question, watch the platform retrieve and reason across different enterprise knowledge types, inspect the evidence behind the answer, understand how uncertainty/conflicts are surfaced, and see how the same architecture can be configured for their organization.

5. Functional Requirements

5.1 Query & answer experience

- Business-oriented query interface with suggested scenario questions.
- Display the interpreted intent and, where appropriate, the planned source/agent path. • Show final answer in an executive-friendly structure: Executive Summary, Key Findings, Supporting Evidence, Data/Metrics, Source Traceability, Confidence/Grounding and Conflicts.
- Clearly distinguish retrieved facts from model-generated inference or synthesis.
- Support follow-up questions while preserving relevant context and access boundaries. **5.2**

Source management

- Upload/register synthetic or public documents with metadata.
- Connect a structured data source such as CSV and/or PostgreSQL.
- Support controlled external sources through allowlisted APIs or approved public sources. • Store source metadata including source type, name, version/date and access classification. • Make it possible to replace an industry's data pack without changing the core application.

5.3 Evidence & traceability

- Every material claim in an answer should be traceable to supporting evidence where available. • Document evidence should identify source and page/section where technically possible. • Structured evidence should identify dataset/table/query or equivalent source context. • External evidence should identify source and retrieval date/time where applicable. • If evidence is insufficient, the system should say so rather than fabricate an answer.

5.4 Conflict handling

- Detect when two or more sources provide materially different values or statements. • Present the conflicting evidence side by side.
- Do not invent a resolution. Explain a reason for the discrepancy only when supported by evidence such as source version/date.
- Allow the final response to state that the evidence is conflicting or unresolved.

6. Multi-Agent Architecture

The source case study describes specialized agents that handle different information sources and combine findings into one contextual response. The new platform should make that architecture visible and reusable. fileciteturn1file0

Query Router / Planner	Understand intent and create execution plan	Identify likely source types, required agents and query path; avoid unnecessary tool calls.
Document Intelligence Agent	Retrieve and reason over documents	Return relevant passages with metadata, page/section context and source identifiers.
Structured Data Agent	Query structured data safely	Generate/execute read-only queries within approved schema and access boundaries; return data and query context.
External Intelligence Agent	Retrieve from approved external sources	Use only allowlisted sources/APIs; distinguish external evidence from internal/synthetic evidence.
Evidence / Validation Agent	Check support and conflicts	Assess whether claims are supported, partially supported, conflicting or insufficiently evidenced.
Synthesis Agent	Produce final response	Combine validated findings into an executive-friendly answer

		while preserving attribution and uncertainty.
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Audit / Observability Layer	Record system behavior	Capture query, user/workspace, agents invoked, sources, model/tool metadata, latency, errors and outcome metrics.

Target flow: User Query → Query Router/Planner → Specialized Agents → Retrieval/Data Access → Evidence Validation → Synthesis → Final Answer + Citations, with Audit/Observability across the workflow.

7. Enterprise Security, Governance & Trust

“Governed by Design” and “Built for Enterprise Use” are core principles in the source case study. The new implementation must demonstrate these principles through actual controls where feasible, rather than relying only on architecture diagrams or marketing copy. fileciteturn1file0

7.1 Access control

- Authentication for the application.
- Role-based access control with at least a demonstrable user/admin distinction.
- Workspace/source-level access boundaries where applicable.
- Authorization enforced server-side—not only hidden in the UI.
- Retrieval must respect source access metadata.

7.2 Secrets & data security

- No credentials, API keys or secrets hard-coded in source code.
- Use environment variables or an appropriate secrets mechanism.
- Do not expose credentials, internal prompts or sensitive configuration through logs or UI. • Separate synthetic/demo data from any real client information.
- Do not use NDA-protected client data, screenshots or proprietary documents in the demo. **7.3**

AI-specific security

- Treat retrieved documents as data, not trusted instructions.
- Test prompt injection and instruction-conflict scenarios.
- Prevent retrieval tools from bypassing authorization.
- Restrict SQL/data tools to approved operations and schemas.
- Prevent tool misuse or data exfiltration through agent instructions.
- Test attempts to extract system prompts or secrets.

7.4 Auditability

At minimum, the platform should be able to record: user, timestamp, workspace/industry, query, agents invoked, sources accessed, documents/data retrieved, model/tool metadata, answer, citations, grounding/conflict status, latency and errors.

8. Observability & Evaluation

8.1 Runtime observability

- Total query latency and stage/agent latency.
- Number and types of sources retrieved.

- Agents invoked and tool calls made.
- Error and timeout information.
- Model/token/usage metadata where available.

- Retrieval and answer quality signals.

8.2 Evaluation framework

Create an evaluation set of approximately 20–30 questions per industry pack. The set must include simple retrieval, multi-source questions, structured-data questions, cross-source reasoning, conflicting evidence, insufficient-evidence cases and multi-agent questions.

Answer correctness	Whether the response reaches the correct conclusion from available evidence.
Grounding	Whether material claims are supported by retrieved evidence.
Citation correctness	Whether cited sources actually support the claims they are attached to.
Retrieval quality	Whether relevant evidence is retrieved and irrelevant sources are minimized.
Conflict handling	Whether conflicting evidence is surfaced rather than silently collapsed.
Insufficient evidence	Whether the system refuses/qualifies answers when support is inadequate.
Latency	Observed response time for representative queries.
Failure rate	Tool, retrieval, orchestration and application failures.

Important: Existing case-study metrics such as 4 agents, 47+ documents, 108 queries and 30–60 second response time belong to the previous implementation and must not be reused as new-demo performance claims unless independently measured in this project. fileciteturn1file0

9. Modularity & Configuration Model

The architecture must prove the source case study's “Modular & Reusable” principle: sources, agents and knowledge should be configurable for different use cases. fileciteturn1file0

9.1 Configuration vs. code

Industry name and terminology	Agent framework/orchestration
Synthetic/public datasets	Authentication and authorization
Document collections	Evidence validation logic
Business scenarios and prompts	Audit and observability
Source definitions and metadata	Core retrieval abstractions
Sample CXO questions	Security controls and access enforcement
Evaluation questions and expected evidence	Common UI/application framework
Industry-specific labels / UI content	Shared data contracts and APIs

9.2 Suggested configuration structure

A clean logical structure may follow a pattern such as:

```
/config/industries/manufacturing/  
industry.yaml  
terminology.yaml  
agents.yaml  
sources.yaml  
scenarios.yaml
```

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```
prompts.yaml  
evaluation.yaml
```

The exact technology and file structure may vary, but the principle is mandatory: adding or replacing an industry pack should require configuration/data changes rather than rewriting the core platform.

10. Industry Strategy

Manufacturing / Industrial	First complete implementation	Production, quality, maintenance, suppliers, procurement, inventory, engineering, safety
Retail & CPG	Second validation pack	Sales, promotions, products, consumer research, inventory, market intelligence
Technology & Services	Future pack / optional validation	Projects, service operations, product docs, knowledge bases, delivery metrics
Insurance	Future pack	Policies, claims, underwriting, regulatory knowledge
Healthcare & Life Sciences	Future pack	Research, SOPs, clinical/operational knowledge, controlled documentation
Financial Services	Future pack	Risk, regulatory, financial and operational knowledge

Manufacturing is selected first for this demo strategy because it offers rich cross-source business questions and strong visual/operational storytelling while avoiding the first demonstration becoming primarily a jurisdiction-specific regulatory compliance exercise. This does not mean the architecture is unsuitable for regulated industries; those should be added as configurations with appropriate security, governance and deployment requirements when pursued.

11. UX & CXO Demo Requirements

11.1 Executive-first interface

- Clean enterprise UI with minimal technical terminology.
- Visible source/evidence panel rather than citations hidden behind technical logs.
- Agent activity should be understandable in business terms—for example, “Reviewed production data” or “Checked maintenance reports”—with technical details available on demand.
- Show confidence/grounding and conflicts without implying mathematical certainty where none exists.
- Include industry/scenario selector so the reusability story can be demonstrated.

11.2 Recommended 3–5 minute demo flow

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0:00–0:20	Show the business problem: critical knowledge is fragmented across documents, systems and external information.
0:20–0:40	Ask a realistic manufacturing CXO question that requires multiple source types.
0:40–1:30	Show the platform identifying the relevant knowledge sources and specialized agents.
1:30–2:15	Show the consolidated answer with key findings and metrics.
2:15–2:45	Open evidence/source traceability and show where important claims came from.
2:45–3:15	Demonstrate a conflict or insufficient-evidence scenario and how the platform handles it.
3:15–3:45	Switch to Retail & CPG configuration or show the configuration layer to prove reuse.

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3:45–5:00	Explain how the same architecture can be adapted to the prospect's sources, terminology and business questions.

12. AI-Assisted Development Policy

AI-assisted coding is explicitly allowed and encouraged where it accelerates development. The intern remains responsible for understanding and owning the resulting implementation.

- Permitted: boilerplate, UI scaffolding, API scaffolding, test generation, documentation, refactoring, debugging assistance and implementation suggestions.
- Required: understand generated code before committing it; review dependencies; verify security implications; test failure cases; and document meaningful architectural decisions.
- Never acceptable: copying code without understanding it, hard-coding credentials, disabling security controls to make a demo work, accepting AI-generated SQL/tool access without validation, or assuming generated tests prove correctness.
- Security review: specifically test prompt injection, tool abuse, authorization bypass, secret exposure, unsafe SQL, malicious documents and data exfiltration.
- Code quality: keep modules small, interfaces clear, configuration separate from core logic, and remove unnecessary AI-generated complexity.

13. Testing Requirements

Unit tests	Core agent logic, retrieval utilities, validation logic, access-control checks and important data transformations.
Integration tests	End-to-end source retrieval, orchestration, evidence flow and answer generation.
Security tests	Authentication, authorization, prompt injection, malicious document behavior, tool restrictions, secret exposure and unsafe data access.
Evaluation tests	Industry evaluation set covering correctness, grounding, citations, conflicts, insufficient evidence and multi-source reasoning.
Regression tests	Previously passing critical scenarios remain passing after changes.
Demo smoke test	Repeatable clean startup and execution of the full demo path.

14. Project Phases & Milestones

1. Architecture & planning	Architecture diagram, component boundaries, data flow, security assumptions, configuration-vs-code map, first industry proposal and CXO storyboard.
2. Core platform	Application shell, authentication, source management, configuration framework and basic query flow.
3. Multi-agent intelligence	Router/planner, document agent, structured-data agent, controlled external agent, evidence validation and synthesis.
4. Enterprise controls	RBAC, source/workspace access, audit logging, observability and security test cases.
5. Manufacturing industry pack	Synthetic/public datasets, documents, scenarios, terminology, evaluation set and polished demo flow.
6. Second-industry validation	Retail & CPG pack demonstrating that the core platform does not require a rebuild.

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7. Evaluation & hardening	Evaluation results, regression suite, security review, latency observations and cleanup.
8. Sales packaging	3–5 minute demo video, screenshots, architecture diagram, demo script and short technical overview.

14.1 Mandatory pre-coding submission

- Architecture diagram.
- Configuration-vs-code boundary.
- Manufacturing data/source plan.
- CXO demo storyboard.
- Agent responsibilities and tool boundaries.
- Production-architecture vs demo-only gap assessment.
- Security threat checklist and evaluation approach.

15. Deliverables

- Working web application demonstrating Enterprise Knowledge Intelligence.
- Manufacturing / Industrial industry pack.
- Retail & CPG validation pack.
- Multi-agent orchestration and source-specific agents.
- Document, structured-data and controlled external retrieval.
- Evidence validation, source traceability, conflict handling and insufficient-evidence behavior.

Authentication, RBAC, access enforcement, audit and observability.

- Automated tests and industry evaluation datasets/results.
- Configuration-driven industry architecture and documentation.
- Architecture diagram and technical overview.
- 3–5 minute sales demo video and demo script.
- Screenshots / visual collateral suitable for a prospect case study.
- Setup, environment and runbook documentation.

16. Definition of Done

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Functionality	A realistic multi-source business question can be answered end-to-end.
Evidence	Important answer claims can be traced to supporting sources; unsupported answers are qualified.
Multi-agent	Specialized agents are actually used for appropriate source types and their work can be observed.
Security	Authentication, authorization and tool/source boundaries are enforced server-side and tested.
Governance	Queries, agents, sources and key execution metadata are auditable.
Modularity	Manufacturing and Retail & CPG use the same core platform with configuration/data changes rather than a rebuild.
Quality	Unit/integration/security/evaluation tests exist and critical scenarios pass.
Performance	New-demo latency is measured and reported rather than borrowed from the previous case study.
UX	A non-technical executive can understand the value and evidence flow quickly.
Sales readiness	The demo tells a business story and clearly shows how the platform could map to a prospect's environment.

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17. Risks & Guardrails

Becoming a generic chatbot	Require multi-source, multi-agent business questions and visible evidence.
Overbuilding infrastructure	Build production-grade architecture and controls, but only infrastructure necessary for the demonstration. Every component should have a clean path to production.
Hard-coded industry logic	Enforce configuration-vs-code review before implementation.
AI-generated code quality	Require human review, tests, security checks and ownership of all committed code.
NDA/client leakage	Use only synthetic or public data; no client identifiers, documents, screenshots or proprietary configurations.
Hallucinated answers	Evidence validation, source traceability and insufficient-evidence handling.
False performance claims	Measure new-demo metrics independently.
Demo optimized but architecture weak	Keep service boundaries, authorization, logging, data contracts and configuration modular from the start.
Regulated-industry distraction	Start with Manufacturing; treat regulated industries as future configurations with their own security/deployment requirements.

18. Recommended Project Brief for the Intern

Build a reusable Enterprise Knowledge Intelligence Reference Platform for Nebula9. Do not build a generic chatbot and do not reproduce any client implementation. Build a sanitized demonstration of how an enterprise can use multiple AI agents to find, validate and synthesize knowledge from different systems.

The first complete scenario will be Manufacturing / Industrial. The platform should combine documents, structured business data and controlled external information, then answer complex business questions with evidence and source traceability. The application must be modular: if the Manufacturing data is replaced with Retail & CPG data, the core application should continue to work without being rebuilt.

Build with production-grade architecture and enterprise controls, but remember that the first release is a demonstration platform. Do not build infrastructure we do not need for the demo. Every component should have a clean path to production.

You may use AI coding tools extensively. However, you are responsible for understanding, reviewing, testing and securing the code you commit. AI-generated code is an accelerator, not a substitute for engineering judgment.

Before writing significant code, submit the architecture, configuration-vs-code map, Manufacturing scenario/data plan, CXO demo storyboard and production-readiness gap assessment for review.

19. Final Product Statement

The final platform should make one idea immediately clear:

Enterprise knowledge does not have to remain fragmented across documents, databases and external sources. A governed, multi-agent knowledge intelligence layer can bring those sources together, retrieve the relevant evidence, reason across them and return a traceable answer in a form business leaders can use.

The implementation should demonstrate that idea credibly, safely and repeatedly across industries—not merely describe it.

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Appendix A — Reference Architecture View

The following logical architecture should guide implementation. Technology choices can vary, but the responsibilities and boundaries should remain clear.

Experience	CXO query UI, answer view, evidence panel, agent activity, industry selector, admin/source management
Application	API layer, session/context management, scenario handling, configuration loader
Orchestration	Query router/planner, agent execution, tool routing, workflow state
Intelligence	Document agent, structured-data agent, external intelligence agent, evidence/validation agent, synthesis agent
Knowledge / Retrieval	Document index/vector retrieval, metadata filters, structured database access, external source adapters
Governance	Authentication, RBAC, source/workspace permissions, audit trail, policy enforcement
Observability	Execution logs, agent/tool traces, latency, errors, evaluation telemetry
Configuration	Industry packs, sources, terminology, scenarios, prompts and evaluation datasets

Appendix B — Example Manufacturing Demo Questions

- What are the major factors contributing to production downtime at Plant A, and what actions have previously reduced similar downtime?
- Which production lines show a combination of rising downtime and declining quality, and what evidence explains the pattern?
- Which suppliers are associated with the highest quality-impact events, and what do supplier documents say about corrective actions?
- What changed between the previous and current maintenance procedures, and is there evidence that the change affected downtime?
- Where do the production report and maintenance documentation disagree, and which source is more recent?

Appendix C — Example Retail & CPG Validation Questions

- Why did Product X underperform in Region A despite higher promotional activity? • Which products show declining sales alongside inventory pressure, and what supporting evidence is available?
- What do consumer research documents indicate about the reasons for the performance change? • Are there conflicts between sales data and market research, and how should the answer qualify them?

Source note: The attached "Multi-Agent RAG Case Study" describes the prior capability as industry-agnostic, AI-powered, reusable across domains, evidence-backed/source-traceable, modular and designed for complex enterprise knowledge environments. Its published historical metrics are treated here as reference context only and are not automatically transferred to this new project.

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